

Multiple Choice (4 marks)

1. Statement 1 | If R is a ring and $f(x)$ and $g(x)$ are in $R[x]$, then $\deg(f(x)+g(x)) = \max(\deg f(x), \deg g(x))$.
(a) True, True
(b) False, False
(c) True, False
(d) False, True
2. Find the sum of the given polynomials in the given polynomial ring. $f(x) = 4x - 5$, $g(x) = 2x^2 - 4x$
(a) $2x^2 + 5$
(b) $6x^2 + 4x + 6$
(c) 0
(d) $x^2 + 1$

True / False (16 marks)

Answer True or False and justify briefly.

1. True or False: The correct answer to 'Let C be the circle in the yz -plane whose equation is $(y - 3)^2 + (z - 4)^2 = 25$. Find the radius of C ' is 5.
2. True or False: The correct answer to 'Find the order of the factor group $(\mathbb{Z}_{11} \times \mathbb{Z}_{15}) / \langle (1, 1) \rangle$ ' is 15.
3. True or False: The correct answer to 'Statement 1 | For any two groups G and G' , there exists a homomorphism from G to G' ' is True.
4. True or False: The correct answer to 'Using Fermat's theorem, find the remainder of 3^{47} when divided by 17' is 9.
5. True or False: The correct answer to 'Determine whether the polynomial in $\mathbb{Z}[x]$ satisfies an Eisenstein criterion' is No.
6. True or False: The correct answer to 'What is the area of an equilateral triangle whose inscribed circle has radius 1' is $\sqrt{3}$.
7. True or False: The correct answer to 'A tree is a connected graph with no cycles. How many non-isomorphic trees with 5 vertices are there?' is 3.
8. True or False: The correct answer to 'Statement 1 | If H is a subgroup of G and a belongs to G then $aH = Ha$ ' is True.

Long Form Response (10 marks)

Provide thorough reasoning and reference key steps.

1. Expand the following expression: $3(8x^2 - 2x + 1)$.

End of Paper